

Examples.

Emails, text documents, Social media posts, Image files (JPG, GIF) Video files, Sensor data from (IoT) devices. It is not easily processed and analyzed through conventional data tools and methods. It is best managed in nonrelational or NoSQL databases or in data lakes which are designed to handle massive amounts of raw data in any format.

Semi-structured data

It is information that does not reside in a relational database but it has some organizational properties that make it easier to analyze. It exists to ease space eg XML data.

6. Data modeling is the process of creating a visual representation of either a whole information system or parts of it to communicate connections between datapoints and structures.

It is important in system design because it provides a structured, visual and standardized way to understand manage and optimize data. It ensures systems are efficient scalable and aligned with business objectives while all maintaining data integrity and supporting future growth.

Ⓐ Conceptual data models → they are domain and offer a high level view of what the system will contain, how it is organized and which business rules are involved.

Ⓑ Logical data models → less abstract and provide greater detail about the concepts and relationships in the domain under consideration.